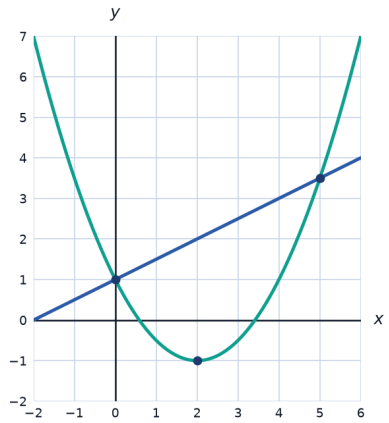
# Interquartile range (Extended)

## Core Concepts: Interquartile range (Extended)

**1** For the model $f(x) = 6x + 5$ in interquartile range (extended), evaluate $f(2)$.




**a.** 17

**b.** 23

**c.** 12

**d.** 21

**2** Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

**a.** $x = 1$ and $x = 4$

**b.** $x = 1$ and $x = 3$

**c.** $x = -1$ and $x = -3$

**d.** $x = 4$ and $x = 3$

**3** A data set recorded for interquartile range (extended) consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

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## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 6x + 5$ in interquartile range (extended), evaluate $f(3)$.

**a.** 23

**b.** 29

**c.** 18

**d.** 27

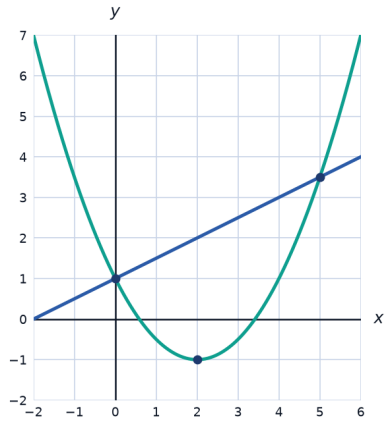
**5** Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

**a.** $x = 2$ and $x = 5$

**b.** $x = 2$ and $x = 4$

**c.** $x = -2$ and $x = -4$

**d.** $x = 6$ and $x = 8$




**6** A data set recorded for interquartile range (extended) consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 6x + 5$ in interquartile range (extended), evaluate $f(4)$.

**a.** 29

**b.** 35

**c.** 24

**d.** 33

**8** Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

**a.** $x = 3$ and $x = 6$

**b.** $x = 3$ and $x = 5$

**c.** $x = -3$ and $x = -5$

**d.** $x = 8$ and $x = 15$

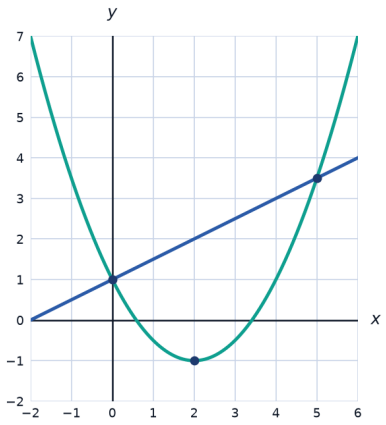
**9** A data set recorded for interquartile range (extended) consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

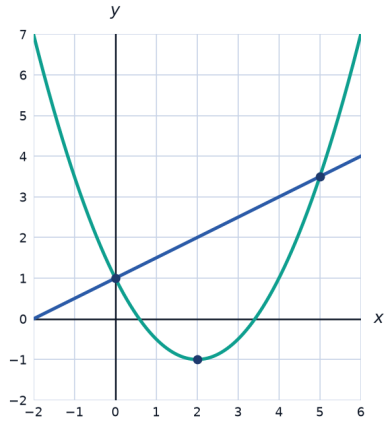
**d.** 22.5




## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 6x + 5$ in interquartile range (extended), evaluate $f(5)$.




**a.** 35

**b.** 41

**c.** 30

**d.** 39

**11** Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

**a.** $x = 4$ and $x = 7$

**b.** $x = 4$ and $x = 6$

**c.** $x = -4$ and $x = -6$

**d.** $x = 10$ and $x = 24$

**12** A data set recorded for interquartile range (extended) consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

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